

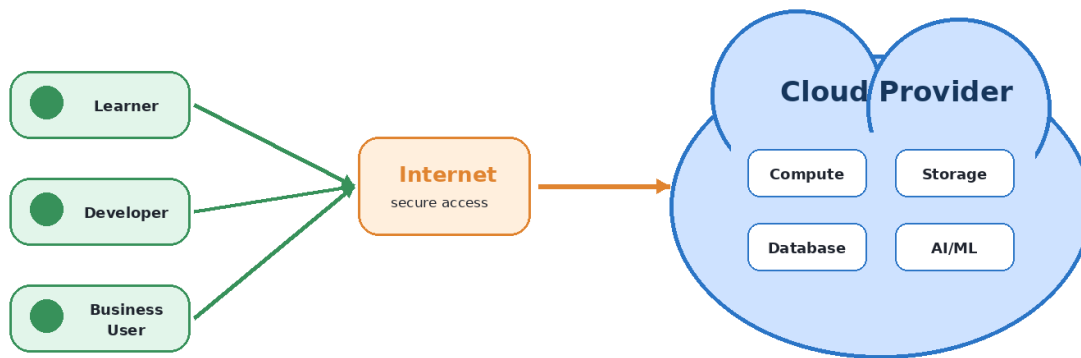
Detailed Tutorial Material

Cloud Computing Fundamentals

Cloud Concepts, Service Models, Deployment Models, and Core Cloud Services Overview

What is Cloud Computing?

IT resources delivered over the internet on demand



Core idea:

- No need to own all hardware
- Create resources in minutes
- Scale when demand changes
- Pay based on actual usage

Cloud Computing Fundamentals - Visual Tutorial

Figure 1: Cloud computing in one visual - users access computing services through the internet.

Purpose of this material

This PDF is designed as a complete 4-hour beginner-friendly tutorial handout. It contains explanations, visual diagrams, comparison tables, trainer notes, mini activities, a final architecture exercise, and quiz questions.

Item	Details
Tutorial title	Cloud Computing Fundamentals
Duration	4 hours
Level	Beginner to foundation level
Audience	Students, freshers, developers, testers, administrators, business analysts, and professionals moving from traditional IT to cloud
Teaching style	Concept -> diagram -> example -> quick check -> activity
Prerequisites	Basic understanding of internet, applications, and databases is helpful but not mandatory

Content Map

Use this content map as the flow for a 4-hour session. The tutorial starts with basic cloud understanding and gradually moves toward service models, deployment models, core cloud services, and a real-world architecture design exercise.

4-Hour Tutorial Flow

Recommended timing for a beginner-friendly session



Trainer rhythm:

Explain concept -> show visual -> give real example -> ask a quick question -> move to next concept.

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Figure 2: Recommended 4-hour tutorial flow.

Module	Time	Main Objective	Teaching Output
1. Introduction to cloud computing	30 min	Explain what cloud computing means and why it matters	Learner can define cloud computing in simple language
2. Key characteristics	25 min	Explain what makes cloud different from traditional IT	Learner understands on-demand, elasticity, measured service, and pooling
3. Service models	45 min	Explain IaaS, PaaS, SaaS, and serverless	Learner can choose the right model for a use case
4. Deployment models	35 min	Explain public, private, hybrid, multi-cloud, and community cloud	Learner understands where workloads can be deployed
5. Core services	60 min	Explain compute, storage, database, networking, IAM, security, monitoring, backup, containers, serverless, AI/analytics	Learner understands common cloud building blocks
6. Reference architecture	35 min	Connect all concepts using an online shopping architecture	Learner can map services to architecture components
7. Hands-on design activity	30 min	Design an online learning platform architecture	Learner applies concepts in a realistic scenario
8. Recap, quiz, and Q&A	20 min	Reinforce learning and clarify doubts	Learner completes quiz and final review

Learning Objectives

By the end of this tutorial, learners should be able to explain cloud computing concepts in simple terms and connect them to real-world application architecture.

- Explain cloud computing and compare it with traditional IT infrastructure.
- Describe the key characteristics of cloud computing: on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service, automation, and global availability.
- Differentiate IaaS, PaaS, SaaS, and serverless computing.
- Differentiate public cloud, private cloud, hybrid cloud, multi-cloud, and community cloud.
- Identify core cloud services: compute, storage, database, networking, IAM, security, monitoring, backup, disaster recovery, containers, Kubernetes, serverless, AI/ML, and analytics.
- Explain a simple real-world architecture using cloud services.
- Participate in an architecture design exercise and justify service choices.

Module 1: Introduction to Cloud Computing

Cloud computing means using computing resources such as servers, storage, databases, networking, software, analytics, and application platforms over the internet instead of owning and managing all infrastructure physically. The core shift is from buying and operating everything yourself to consuming technology as a service.

1.1 What is Cloud Computing?

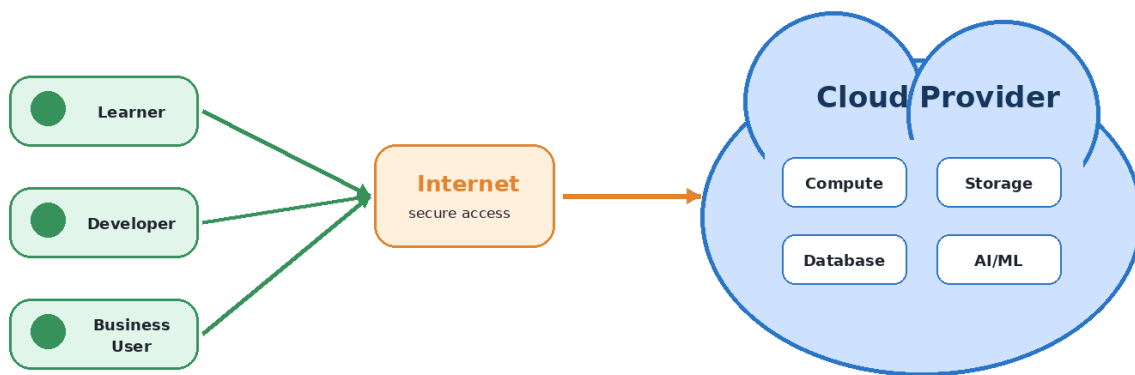
Learning outcome

Learners should be able to define cloud computing in one simple sentence.

Cloud computing is the delivery of computing resources over the internet on demand. A learner can think of it like electricity: instead of building your own power plant, you consume power from a provider and pay based on usage. Similarly, in cloud computing, you consume IT capacity from a cloud provider and pay based on what you use.

What is Cloud Computing?

IT resources delivered over the internet on demand



Core idea:

- No need to own all hardware
- Create resources in minutes
- Scale when demand changes
- Pay based on actual usage

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Figure 3: Cloud computing delivers compute, storage, database, networking, and other services over the internet.

Trainer explanation points

- Cloud computing is not only about servers; it includes platforms, databases, AI services, security, monitoring, and business applications.
- The customer uses a cloud console, API, CLI, or automation pipeline to create and manage services.
- The provider operates global data centers and offers services that customers consume on demand.

Simple example

A small company wants to launch a website. Instead of buying a physical server, it can create a cloud virtual machine or app service, attach storage, configure networking, and go live much faster.

1.2 Traditional IT vs Cloud IT

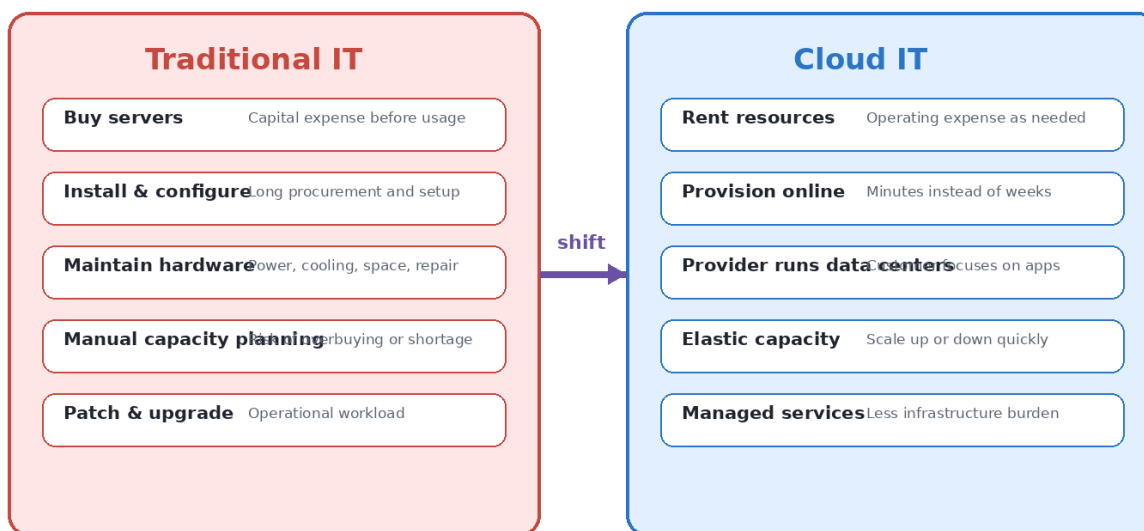
Learning outcome

Learners should understand why organizations move from physical infrastructure to cloud services.

In traditional IT, a company usually purchases hardware, sets up data centers, installs operating systems, configures networks, maintains storage, and handles replacement cycles. In cloud IT, many of these tasks shift to the cloud provider, allowing the organization to focus more on applications, data, security configuration, and business value.

Traditional IT vs Cloud IT

Same business need, different operating model



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Figure 4: Traditional IT requires more ownership and maintenance; cloud IT provides faster consumption and elasticity.

Trainer explanation points

- Traditional IT often requires high upfront capital investment.
- Cloud IT uses an operating expense model where teams consume resources as needed.
- Cloud does not remove responsibility; it changes the responsibility boundary.

Simple example

If a retail website expects high traffic only during a festival sale, cloud allows capacity to increase temporarily and decrease after the sale. In traditional IT, the company may have to buy extra servers that remain underused later.

Mini activity

Ask learners: What are three problems a company faces when it owns all infrastructure itself? Expected answers: cost, slow setup, maintenance, scaling, security operations, disaster recovery.

1.3 Why Businesses Adopt Cloud

Organizations adopt cloud computing because it supports speed, flexibility, scalability, innovation, and cost visibility. Cloud also gives access to advanced capabilities such as AI, machine learning, IoT, data analytics, automation, and globally distributed infrastructure.

Business Need	Cloud Benefit	Example
Faster project launch	Provision infrastructure and platforms quickly	Create a development environment in minutes
Variable demand	Scale resources up and down	Increase capacity during online sale
Lower upfront investment	Avoid large hardware purchase at the start	Use pay-as-you-go virtual servers
Global reach	Deploy apps closer to users	Serve customers across countries
Modern innovation	Use AI, analytics, IoT, and automation services	Build recommendation, forecasting, or chatbot features
Business continuity	Use backup, replication, and disaster recovery	Recover service after regional outage

Trainer note

Avoid presenting cloud as automatically cheaper. Cloud can reduce upfront cost and improve flexibility, but without governance, tagging, right-sizing, and monitoring, cloud bills can grow quickly.

Module 2: Key Characteristics of Cloud Computing

Cloud computing has a few common characteristics that make it different from traditional IT. These characteristics help explain why cloud is useful for modern digital systems.

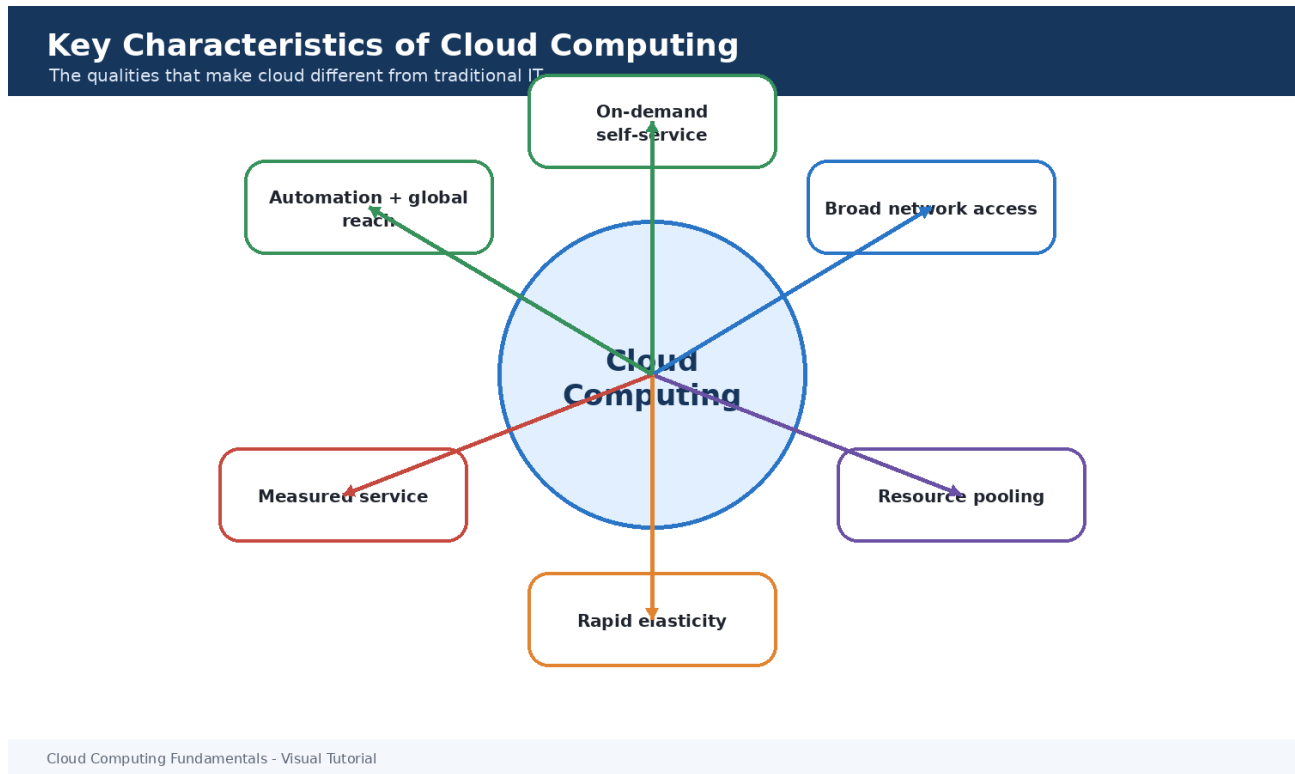


Figure 5: Six important characteristics that define the cloud operating model.

2.1 On-Demand Self-Service

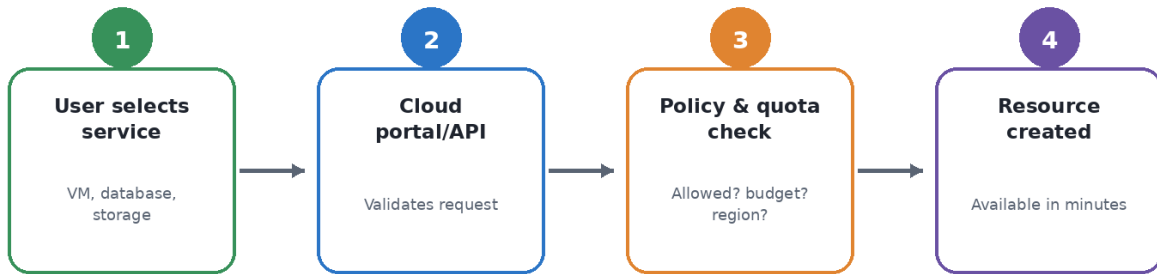
Learning outcome

Learners should understand that authorized users can provision services when needed.

On-demand self-service means a user or application can request cloud resources without waiting for a manual infrastructure process every time. This does not mean anyone can create anything. In a well-managed cloud environment, self-service is controlled by identity, permissions, policies, quotas, budgets, approvals, and governance rules.

On-Demand Self-Service

Users can provision approved resources without manual infrastructure tickets



Teaching point

Cloud self-service does not mean uncontrolled access. Mature organizations combine self-service with policies, roles, limits, approvals, tags, and cost controls.

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Figure 6: Self-service flow from user request to resource creation.

Trainer explanation points

- Self-service improves speed and reduces dependency on manual infrastructure tickets.
- Governance remains important: organizations must control who can create what, where, and at what cost.
- Automation makes provisioning repeatable and reduces human error.

Simple example

A developer creates a test database from a cloud console for a project. The database is created within minutes, but only because the developer has the right role and the organization has allowed that service.

2.2 Broad Network Access

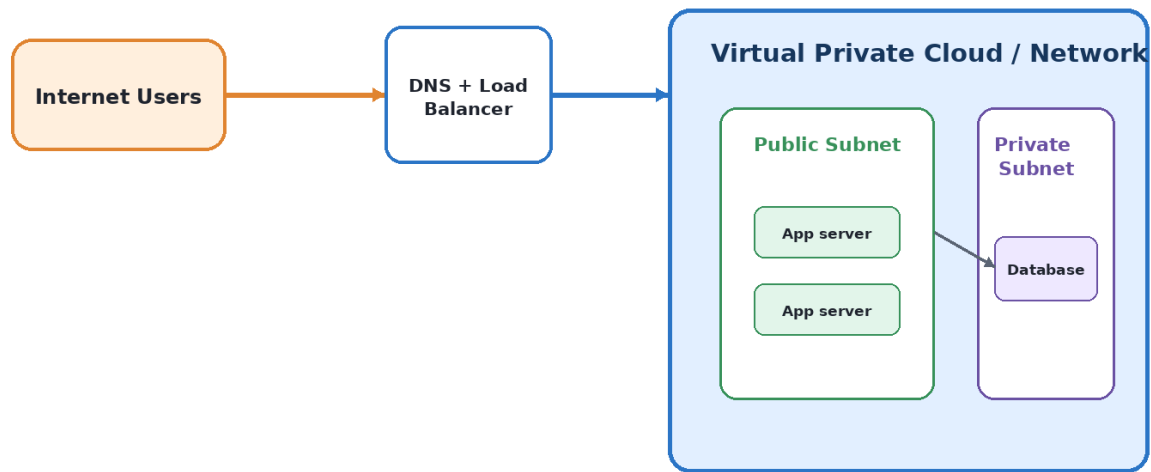
Learning outcome

Learners should understand that cloud services are accessible over networks through standard mechanisms.

Broad network access means cloud services can be accessed over the internet or private network connections using standard devices and protocols. Users may access SaaS applications from browsers, developers may access APIs from code, and enterprises may connect data centers to cloud platforms through private links or VPNs.

Cloud Networking Services

Connect resources securely and control traffic flow



Key concepts: VPC/VNet, subnet, route table, DNS, load balancer

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Figure 7: Cloud networking allows controlled access from users to application and database resources.

Trainer explanation points

- Cloud access must be secured using authentication, authorization, encryption, and network controls.
- Not every cloud resource should be public. Databases are usually placed in private networks.
- Good architecture separates public entry points from private backend resources.

Simple example

A customer accesses an e-commerce website over the internet, but the database remains private and is accessible only to backend application servers.

2.3 Resource Pooling

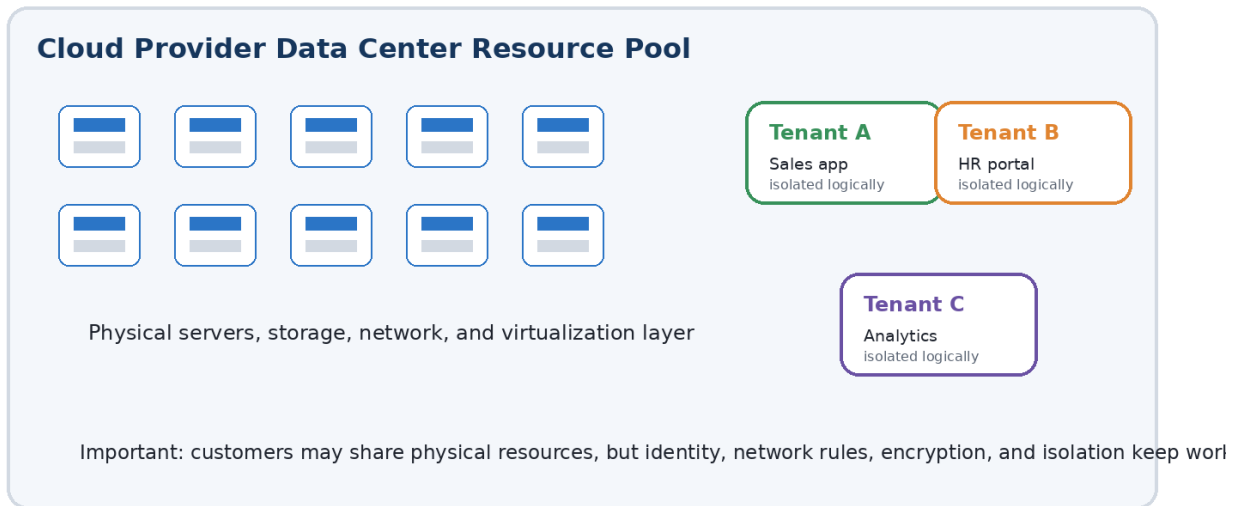
Learning outcome

Learners should understand how cloud providers share large pools of infrastructure across many customers securely.

Resource pooling means cloud providers operate large pools of compute, storage, and networking capacity that can be shared among multiple customers. The customers do not usually know the exact physical server being used. The provider uses virtualization, isolation, security controls, and scheduling to allocate resources efficiently.

Resource Pooling and Multi-Tenancy

Shared infrastructure, logically separated customers



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Figure 8: Multiple tenants use a shared provider resource pool while staying logically isolated.

Trainer explanation points

- Resource pooling enables efficiency and lower cost because the provider can optimize utilization.
- Cloud customers are logically isolated even if the underlying physical infrastructure is shared.
- Security design must include identity, network isolation, encryption, and monitoring.

Simple example

A cloud provider may run workloads from many companies in the same data center, but each customer sees only its own resources through its cloud account and permissions.

2.4 Rapid Elasticity

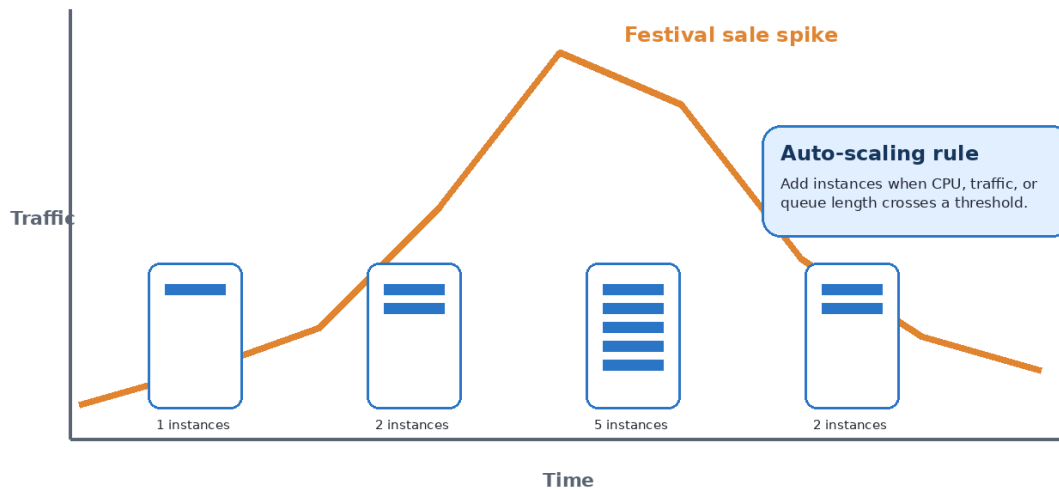
Learning outcome

Learners should understand how cloud capacity can expand and shrink based on demand.

Rapid elasticity means resources can be scaled up or down quickly. A system can add more instances during traffic spikes and remove them when traffic decreases. Elasticity is one of the most important reasons cloud is used for web applications, streaming platforms, analytics workloads, and seasonal business demand.

Rapid Elasticity and Auto-Scaling

Capacity expands and contracts with demand



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Figure 9: Auto-scaling adds instances during demand spikes and reduces them when demand falls.

Trainer explanation points

- Elasticity is useful only when architecture supports scaling. Poorly designed applications may not scale automatically.
- Scaling can be vertical, such as increasing CPU or memory, or horizontal, such as adding more application instances.
- Auto-scaling should be combined with monitoring and cost controls.

Simple example

A ticket booking app may need extra capacity when popular event tickets go live. Auto-scaling allows the app to add capacity for that period and reduce it after demand normalizes.

2.5 Measured Service

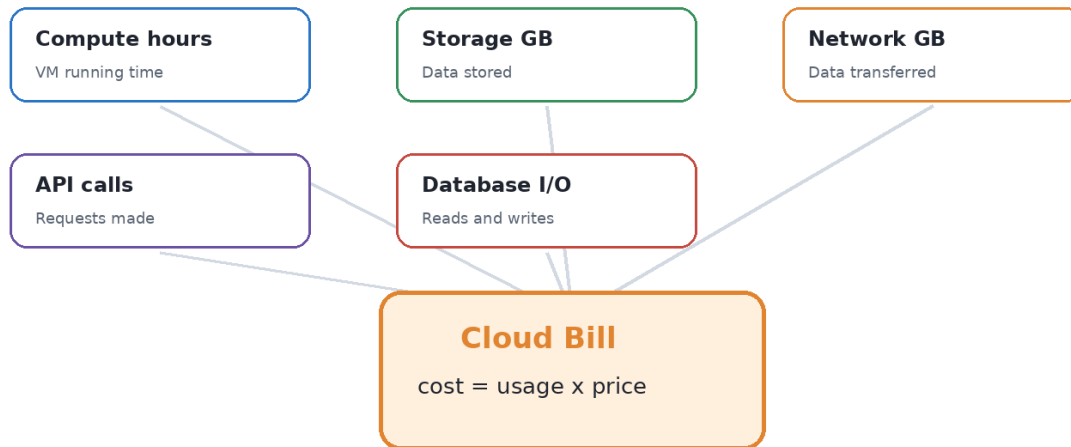
Learning outcome

Learners should understand that cloud usage is metered and billed based on consumption.

Measured service means the cloud provider tracks usage such as compute hours, storage volume, data transfer, API calls, and database operations. This makes billing transparent, but it also requires cost management discipline. Teams should tag resources, set budgets, monitor usage, and clean up unused services.

Measured Service and Pay-As-You-Use

Usage is metered and converted into cost reports



Teaching point: measured service enables cost visibility, but teams still need tagging, budgets, alerts

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Figure 10: Different usage meters contribute to the final cloud bill.

Trainer explanation points

- Measured service supports pay-as-you-go billing.
- Cost can be optimized by right-sizing, shutting down idle resources, selecting correct storage tier, and using reserved capacity where suitable.
- Cost visibility is a technical and governance responsibility.

Simple example

If a virtual machine runs continuously for a month, the bill will be different from a function that runs only for a few seconds after each file upload.

2.6 Automation, Self-Healing, and Global Availability

Modern cloud platforms support automation through templates, infrastructure as code, APIs, pipelines, and policies. Self-healing means the platform can restart failed components, replace unhealthy instances, or shift traffic based on health checks. Global availability means services can be deployed in multiple regions and zones for lower latency and better resilience.

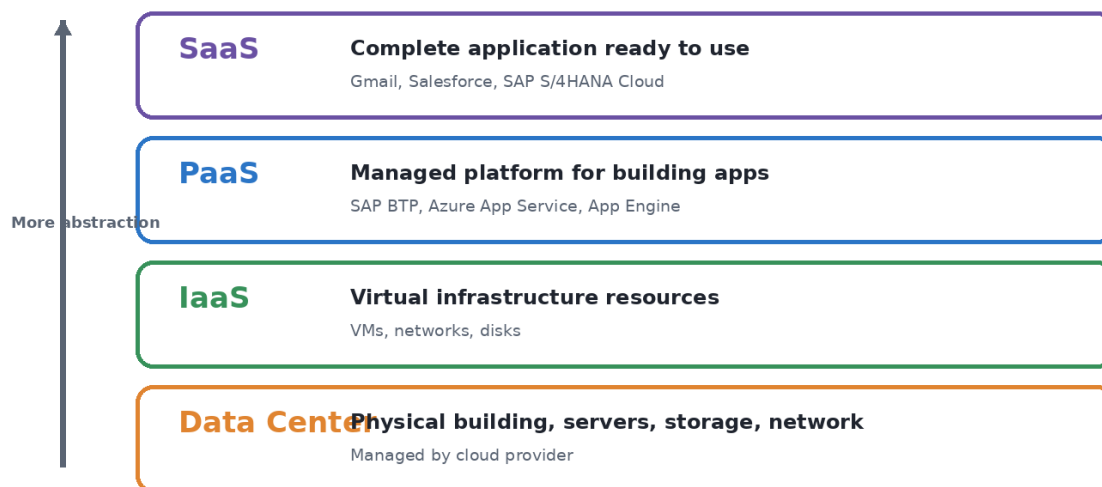
- Automation reduces manual setup and makes environments repeatable.
- Self-healing improves reliability by responding to common failures automatically.
- Global availability helps serve users from nearby regions and improves disaster recovery planning.
- These capabilities require good architecture, monitoring, and operational discipline.

Module 3: Cloud Service Models

Cloud service models define how much responsibility is handled by the cloud provider and how much remains with the customer. The four models covered in this tutorial are IaaS, PaaS, SaaS, and serverless computing.

Cloud Service Models

From raw infrastructure to complete software



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Figure 11: Service models move from infrastructure control to complete software consumption.

Shared Responsibility Model

Who manages what changes by service model

Layer	On-Premise	IaaS	PaaS	SaaS
Data	Customer	Customer	Customer	Customer config
Application	Customer	Customer	Customer	Provider
Runtime	Customer	Customer	Provider	Provider
Middleware	Customer	Customer	Provider	Provider
Operating System	Customer	Customer	Provider	Provider
Virtualization	Customer	Provider	Provider	Provider
Servers	Customer	Provider	Provider	Provider
Storage	Customer	Provider	Provider	Provider
Networking	Customer	Provider	Provider	Provider
Physical Facility	Customer	Provider	Provider	Provider

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Figure 12: Responsibility shifts from customer to provider as the service model becomes more managed.

Simple explanation

The easiest way to explain service models is through a responsibility stack. In IaaS, the customer manages more. In PaaS, the platform is managed by the provider. In SaaS, the complete application is provided. In serverless, the customer focuses mainly on small units of code and events.

3.1 IaaS - Infrastructure as a Service

Learning outcome

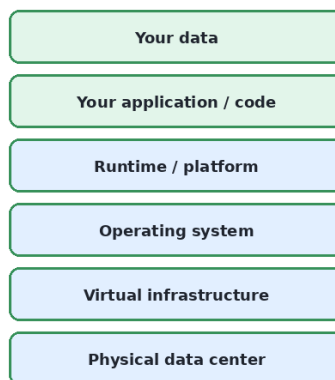
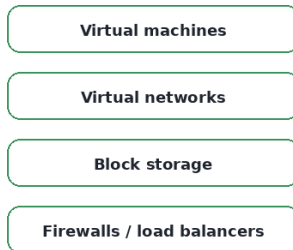
Learners should understand that IaaS provides virtual infrastructure but still requires the customer to manage operating systems and applications.

Infrastructure as a Service provides virtual machines, storage, and networking. It is closest to traditional infrastructure but delivered through cloud. IaaS gives more control than PaaS or SaaS. However, that control also means the customer must manage more layers such as operating system updates, middleware, runtime, application deployment, backup configuration, and security settings.

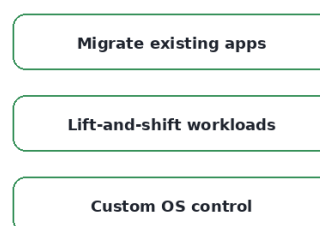
IaaS - Infrastructure as a Service

Customer gets virtual infrastructure and manages OS, runtime, app, and data.

Typical building blocks



Best suited for



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Figure 13: IaaS provides infrastructure building blocks while the customer manages OS, app, and data.

Trainer explanation points

- Use IaaS when you need operating system control, custom networking, or migration of existing applications.
- IaaS is flexible but requires more administration than PaaS or SaaS.
- Security patching of OS and application is usually a customer responsibility.

Simple example

A company has an existing application running on Linux servers. It can migrate the application to cloud virtual machines with minimal code change.

Mini activity

Ask learners to list what the customer manages in IaaS. Expected: OS, middleware, runtime, app, data, access control, configuration.

3.2 PaaS - Platform as a Service

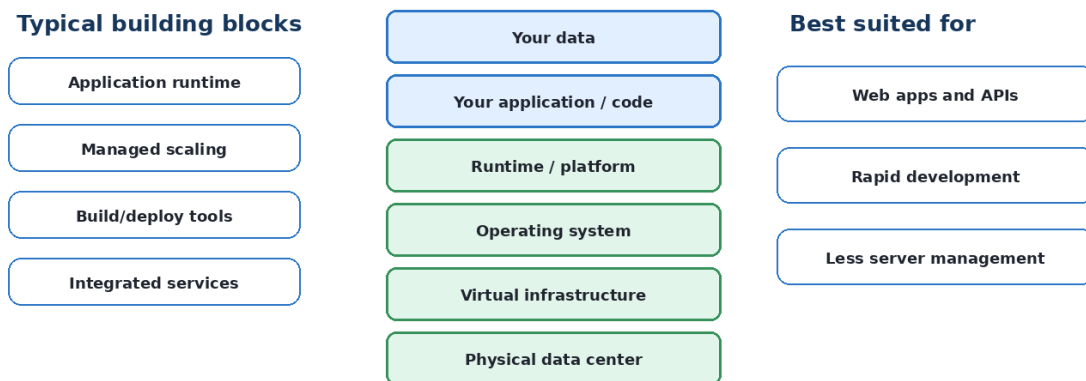
Learning outcome

Learners should understand that PaaS helps developers deploy applications without managing the full server stack.

Platform as a Service provides a managed application platform. Developers deploy code, configure environment variables, connect services, and manage application logic. The provider manages infrastructure, operating system, runtime, scaling platform, and many operational concerns. PaaS is useful when speed of development and reduced operational overhead are important.

PaaS - Platform as a Service

Customer deploys application code on a managed platform.



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Figure 14: PaaS allows developers to focus on application code and data.

Trainer explanation points

- Use PaaS for web apps, APIs, business extensions, and rapid application development.
- PaaS reduces infrastructure management but may reduce low-level control.
- SAP BTP is a strong example of enterprise PaaS for SAP-centric extensions, integration, data, workflow, and AI scenarios.

Simple example

A development team builds an approval workflow app and deploys it on a managed app platform instead of manually creating and patching servers.

3.3 SaaS - Software as a Service

Learning outcome

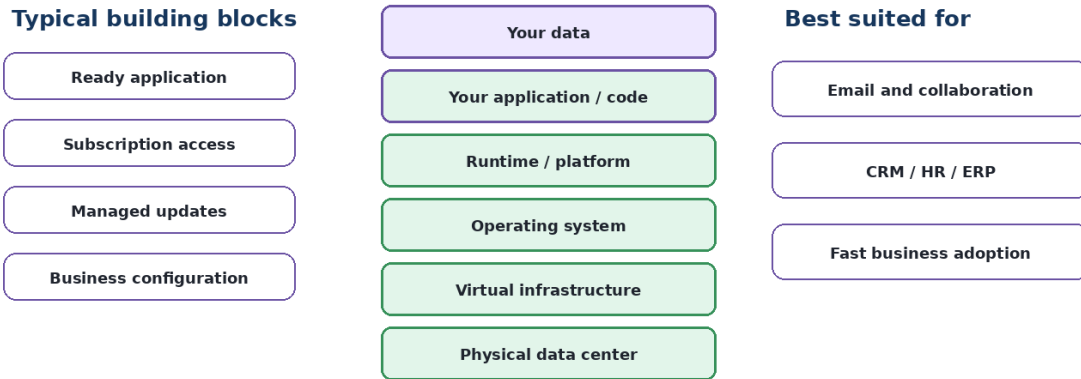
Learners should understand that SaaS is a complete application consumed through subscription or online

access.

Software as a Service provides complete software applications over the internet. The customer does not install or maintain the application infrastructure. The provider manages the application, updates, infrastructure, and platform. The customer manages users, business settings, data usage, and access configuration.

SaaS - Software as a Service

Customer uses a complete application delivered over the internet.



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Figure 15: SaaS provides a ready-to-use application with minimum infrastructure responsibility for the customer.

Trainer explanation points

- Use SaaS when the business need can be met by a standard application.
- SaaS adoption is usually faster than building software from scratch.
- Customization may be limited compared with custom-built applications.

Simple example

A company uses Microsoft 365 for email and collaboration instead of hosting its own mail servers. It manages users and policies, while the provider runs the service.

3.4 Serverless Computing

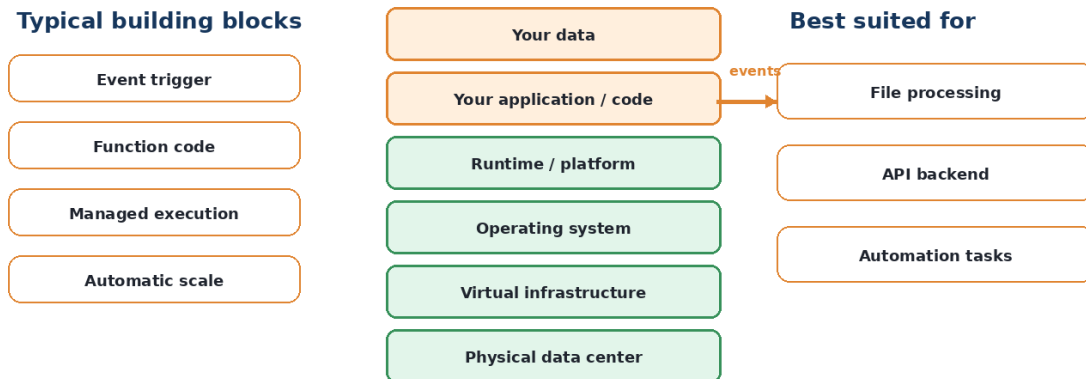
Learning outcome

Learners should understand that serverless lets developers run code based on events without managing servers.

Serverless computing allows code to run when triggered by events such as API requests, file uploads, queue messages, scheduled jobs, or database changes. The provider automatically handles server provisioning, scaling, and execution. The customer focuses on function code, event design, security permissions, and monitoring.

Serverless Computing

Customer writes functions or workflows. Provider runs, scales, and charges per execution.



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Figure 16: Serverless focuses on code and events while the provider manages execution infrastructure.

Trainer explanation points

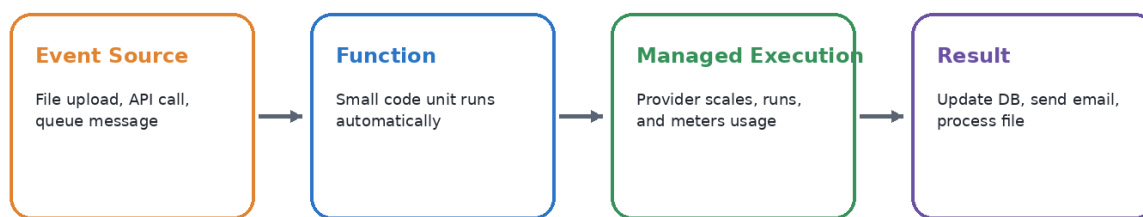
- Serverless is good for event-driven workloads, automation, APIs, and background jobs.
- Serverless can reduce cost when workloads are intermittent.
- It may not be best for long-running processes, workloads requiring full OS control, or extremely predictable always-on workloads.

Simple example

When a user uploads an invoice PDF, a serverless function can extract metadata, store it in a database, and send a notification.

Serverless Event-Driven Flow

Code runs only when an event triggers it



Why it matters: no server provisioning, automatic scale, and cost is often tied to execution count/duration.

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Figure 17: Serverless functions run when events trigger them.

3.5 Service Model Comparison

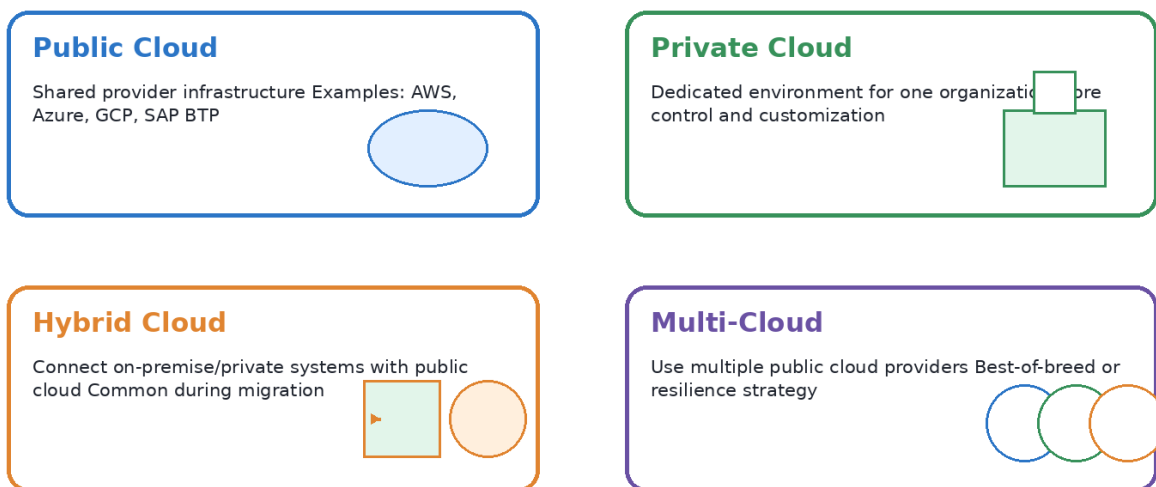
Model	What customer mainly manages	What provider mainly manages	Best for	Example services
IaaS	OS, middleware, runtime, application, data, security configuration	Physical data center, servers, storage, networking, virtualization	Lift-and-shift migration, custom OS control	Amazon EC2, Azure Virtual Machines, Google Compute Engine
PaaS	Application code, business logic, data, app-level security	Infrastructure, OS, runtime, middleware, platform scaling	Web apps, APIs, enterprise extensions	SAP BTP, Azure App Service, Google App Engine, Elastic Beanstalk
SaaS	Users, roles, business configuration, data usage	Complete software stack and updates	Ready business applications	Gmail, Microsoft 365, Salesforce, SAP S/4HANA Cloud
Serverless	Function code, event design, permissions, monitoring	Execution environment, scaling, server management	Event-driven automation and APIs	AWS Lambda, Azure Functions, Google Cloud Functions

Module 4: Cloud Deployment Models

Deployment models describe where cloud infrastructure is located, who owns or controls it, and how the organization uses it. The right deployment model depends on compliance, cost, risk, performance, legacy systems, business strategy, and skills.

Cloud Deployment Models

Where cloud resources live and how organizations consume them



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Figure 18: Public, private, hybrid, and multi-cloud deployment models at a glance.

4.1 Public Cloud

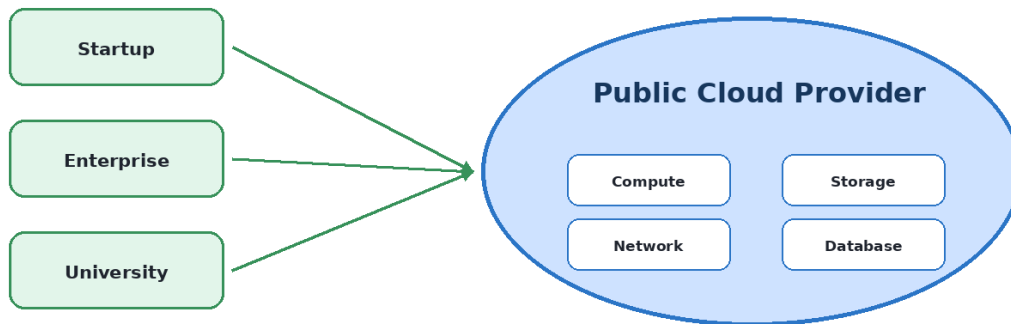
Learning outcome

Learners should understand that public cloud services are delivered by third-party providers on shared infrastructure.

Public cloud is owned and operated by cloud providers and consumed by many customers. It offers quick provisioning, global scale, broad service catalogs, and lower upfront cost. Public cloud is widely used for websites, mobile apps, analytics, AI, SaaS platforms, development environments, and enterprise extensions.

Public Cloud

Cloud services run on infrastructure owned and operated by a provider.



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Figure 19: Public cloud provides shared provider infrastructure to many customers with logical separation.

Trainer explanation points

- Public cloud is not the same as public access. A public cloud resource can still be private and secured.
- It is usually the fastest model for starting new digital projects.
- Governance is essential for identity, network security, compliance, and cost control.

Simple example

A startup hosts its product on public cloud because it can begin small and scale as users grow.

4.2 Private Cloud

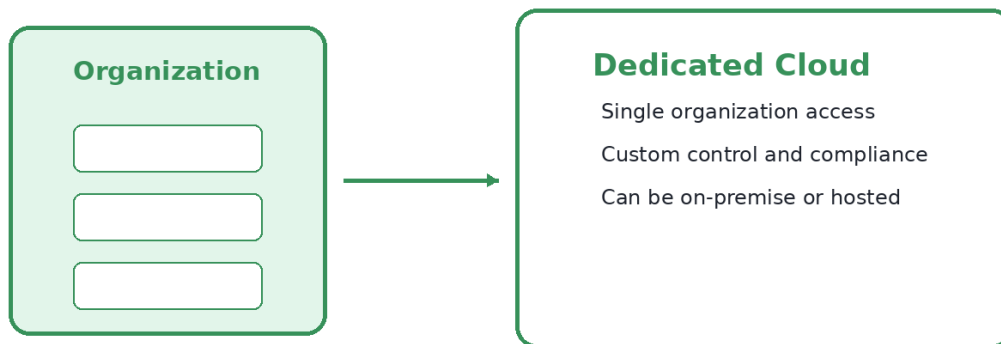
Learning outcome

Learners should understand that private cloud is dedicated to one organization.

Private cloud is a cloud environment used by a single organization. It can be hosted in the company data center or by a third-party provider. It provides more control and customization, but it may require more management effort and cost than public cloud.

Private Cloud

A dedicated cloud environment for one organization.



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Figure 20: Private cloud dedicates a cloud environment to one organization.

Trainer explanation points

- Private cloud is common where regulatory, security, or customization needs are strong.
- It can provide cloud-like automation and self-service inside a dedicated environment.
- The organization must still plan capacity, operations, upgrades, and cost management.

Simple example

A bank may use a private cloud for sensitive workloads that require stricter internal control and compliance.

4.3 Hybrid Cloud

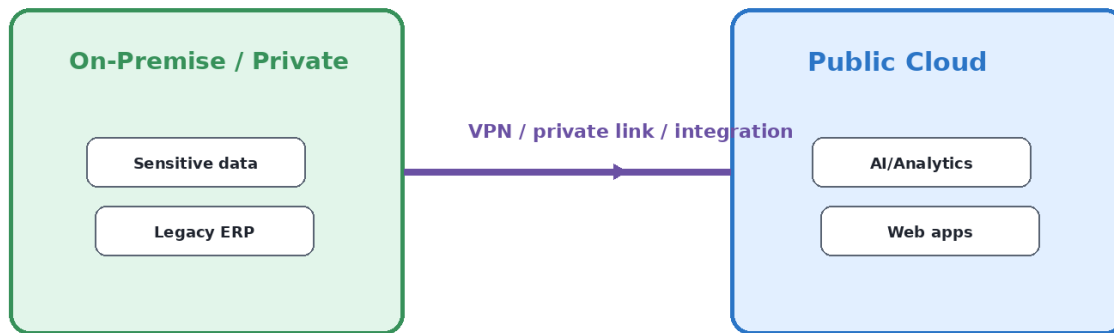
Learning outcome

Learners should understand that hybrid cloud connects private/on-premise systems with public cloud.

Hybrid cloud combines on-premise or private cloud environments with public cloud services. It is common in enterprises that have existing systems but want to use public cloud for innovation, analytics, AI, disaster recovery, or customer-facing applications.

Hybrid Cloud

Public cloud and private/on-premise environments connected together.



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Figure 21: Hybrid cloud connects on-premise/private systems with public cloud services.

Trainer explanation points

- Hybrid cloud is often a practical migration path rather than a one-time replacement.
- Integration, identity federation, network connectivity, and data governance are critical.
- Hybrid cloud can keep sensitive or legacy data on-premise while using public cloud for modern capabilities.

Simple example

A healthcare organization keeps patient records in a private environment but uses public cloud analytics to generate operational reports from approved anonymized data.

4.4 Multi-Cloud

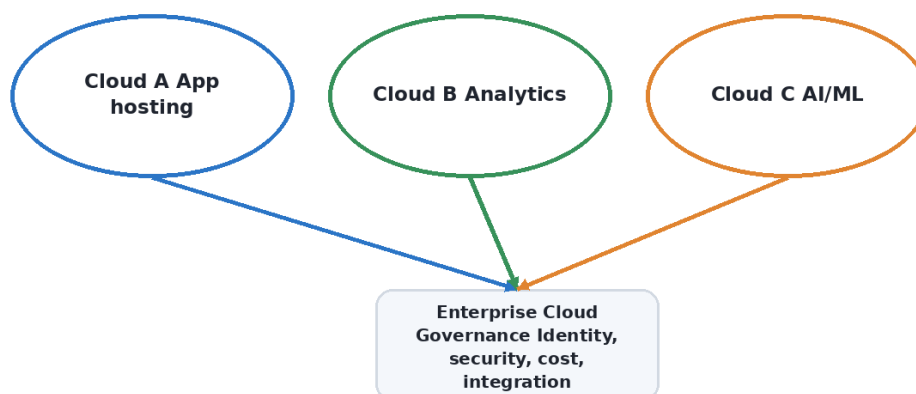
Learning outcome

Learners should understand that multi-cloud means using multiple public cloud providers intentionally.

Multi-cloud means an organization uses services from more than one cloud provider. This may be done to avoid lock-in, use best-of-breed services, meet regional needs, improve resilience, or align with different teams. Multi-cloud can be powerful, but it increases complexity in skills, security, integration, monitoring, and cost governance.

Multi-Cloud

The organization uses two or more cloud providers intentionally.



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Figure 22: Multi-cloud combines services from multiple providers under common governance.

Trainer explanation points

- Multi-cloud is not automatically better; it must have a clear business or technical reason.
- Common governance, identity, networking, monitoring, and cost management become very important.
- Skills and operating standards must be consistent across providers.

Simple example

An enterprise uses one cloud for SAP extensions, another for analytics, and another for AI workloads, while enforcing centralized governance.

4.5 Community Cloud

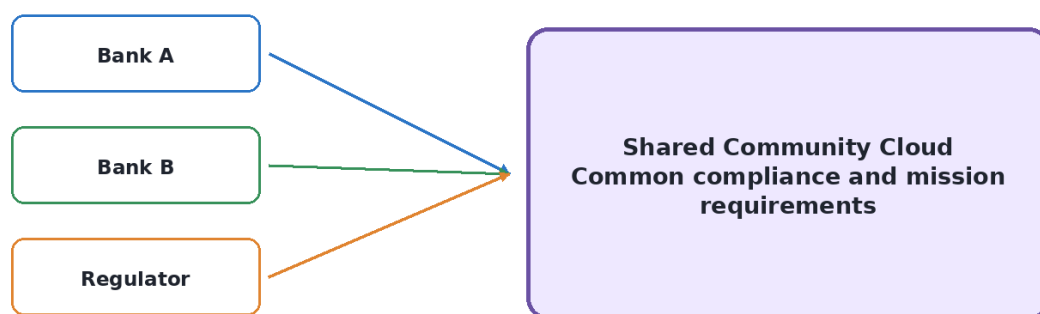
Learning outcome

Learners should understand that community cloud is shared by organizations with similar needs.

Community cloud is a less common model where a cloud environment is shared by organizations with similar regulatory, security, industry, or mission requirements. The model is relevant for sectors such as government, healthcare, education, or financial services consortia.

Community Cloud

Shared by organizations with similar regulatory or mission needs.



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Figure 23: Community cloud serves multiple organizations with common compliance or mission needs.

Trainer explanation points

- Community cloud focuses on shared requirements rather than open public consumption.
- It may be managed by one or more community members or a third-party provider.
- Cost and governance are shared according to the community arrangement.

Simple example

Multiple government departments may use a shared cloud environment built around common security and compliance standards.

4.6 Deployment Model Comparison

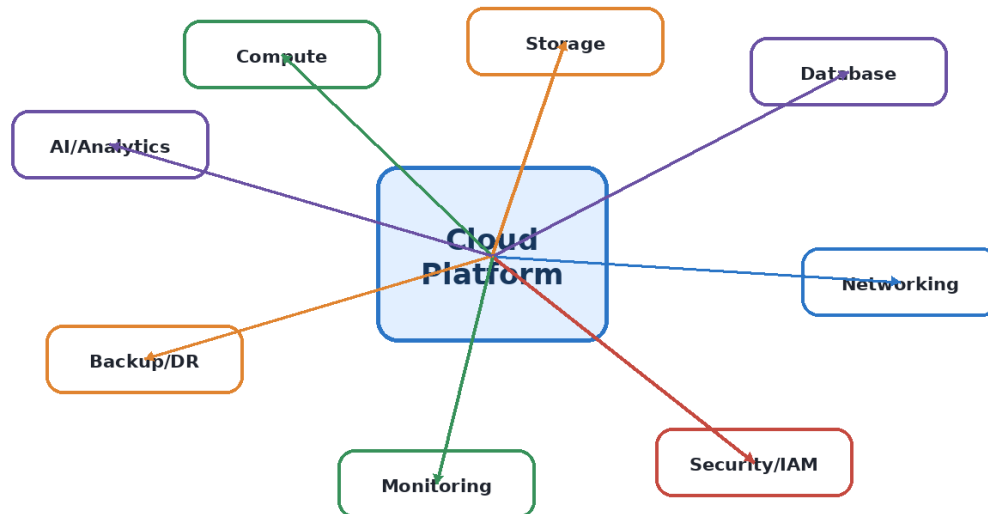
Model	Ownership / Access	Advantages	Challenges	Typical Use Cases
Public cloud	Third-party provider, shared infrastructure, logical isolation	Fast, scalable, global, broad services	Governance, compliance, cost control	Web apps, analytics, AI, development, SaaS
Private cloud	Dedicated to one organization	Control, customization, compliance alignment	Higher management effort and cost	Banking, defense, regulated systems
Hybrid cloud	Combination of on-premise/private and public cloud	Gradual migration, flexibility, sensitive data control	Integration and identity complexity	Enterprise modernization, DR, analytics
Multi-cloud	Multiple public cloud providers	Best-of-breed, resilience, vendor flexibility	Skills, monitoring, security, cost complexity	Large enterprises, global workloads
Community cloud	Shared by similar organizations	Common compliance and mission needs	Governance and ownership complexity	Government, healthcare, education consortia

Module 5: Core Cloud Services Overview

Core cloud services are the building blocks used to design applications. Although each cloud provider has different product names, most cloud platforms offer similar service families.

Core Cloud Services Overview

Most cloud platforms organize capabilities into common service families



Cloud Computing Fundamentals - Visual Tutorial

Figure 24: Core cloud services form the foundation for modern applications.

5.1 Compute Services

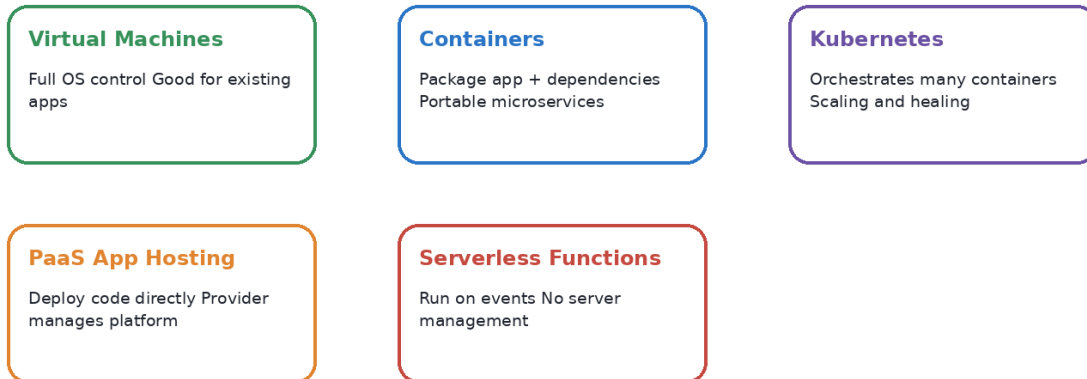
Learning outcome

Learners should understand the different ways cloud can run application workloads.

Compute services provide processing power for applications, APIs, background jobs, analytics workloads, and enterprise systems. The most common compute choices are virtual machines, containers, Kubernetes, PaaS app hosting, and serverless functions. The right choice depends on control, speed, portability, scale, operating effort, and cost pattern.

Compute Services

Processing power to run applications, APIs, jobs, and workloads



Selection rule: choose based on control needed, operational effort, scalability, portability, and cost pattern.

Cloud Computing Fundamentals - Visual Tutorial

Figure 25: Cloud compute options range from virtual machines to serverless functions.

Trainer explanation points

- Virtual machines provide control but require more administration.
- Containers package applications and dependencies for portability.
- Kubernetes manages container scheduling, scaling, and recovery.
- PaaS app hosting reduces server management.
- Serverless runs event-driven code without server provisioning.

Simple example

For an old application needing a specific OS, use VMs. For a modern microservice app, use containers or Kubernetes. For simple event processing, use serverless.

5.2 Storage Services

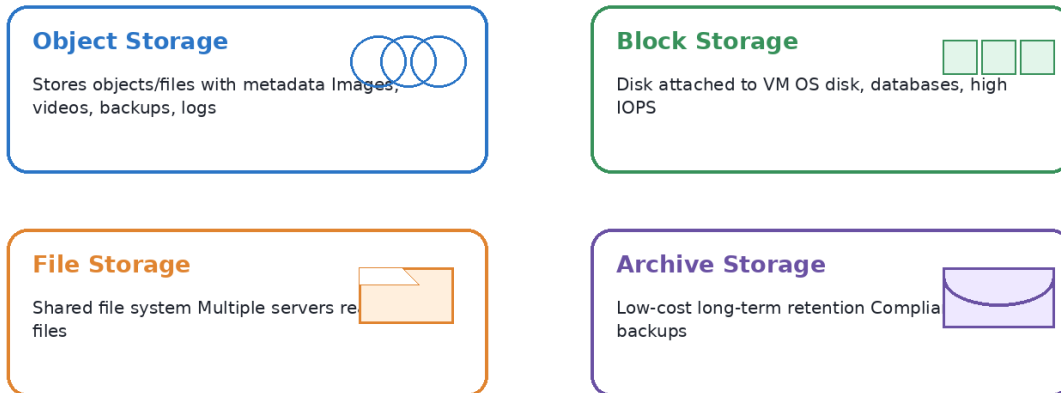
Learning outcome

Learners should understand how object, block, file, and archive storage differ.

Storage services hold data such as files, images, backups, logs, videos, documents, database disks, and long-term archives. Object storage is common for files and unstructured data. Block storage is used like a disk for virtual machines. File storage provides shared file systems. Archive storage keeps rarely accessed data at low cost.

Cloud Storage Services

Different storage types solve different data access patterns



Cloud Computing Fundamentals - Visual Tutorial

Figure 26: Cloud storage types are selected based on access pattern and workload need.

Trainer explanation points

- Object storage is excellent for images, documents, backups, data lakes, logs, and media files.
- Block storage is usually attached to virtual machines and can support high-performance workloads.
- File storage supports shared access by multiple compute instances.
- Archive storage reduces cost for long-retention data but may have slower retrieval.

Simple example

An e-commerce app stores product images in object storage, application logs in object storage, database disks in block storage, and yearly compliance backups in archive storage.

5.3 Database Services

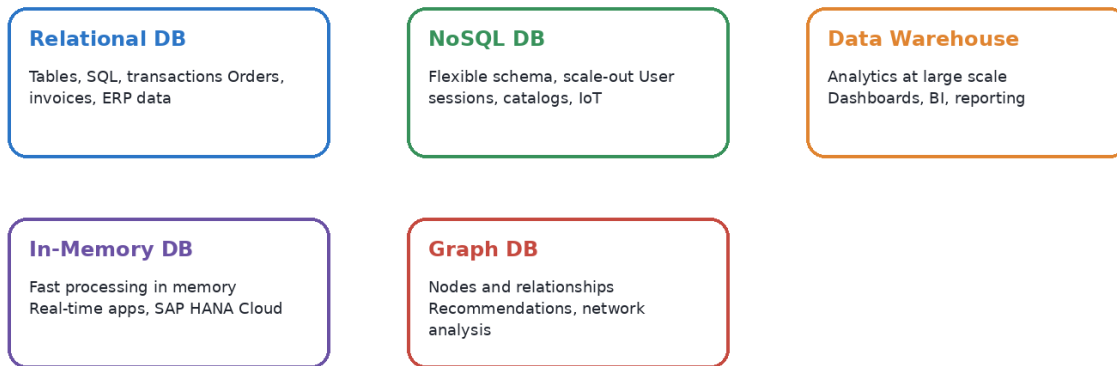
Learning outcome

Learners should understand why different database types exist in cloud platforms.

Database services store structured and unstructured information. Cloud providers offer managed relational databases, NoSQL databases, data warehouses, in-memory databases, graph databases, and caching services. Managed databases reduce operational tasks such as patching, backups, replication, and monitoring.

Cloud Database Services

Managed engines for structured, semi-structured, and analytical data



Cloud Computing Fundamentals - Visual Tutorial

Figure 27: Cloud database services support transactional, analytical, flexible, in-memory, and graph data patterns.

Trainer explanation points

- Relational databases are good for transactions and structured tables.
- NoSQL databases support flexible schema and high-scale access patterns.
- Data warehouses are optimized for analytical reporting.
- In-memory databases support fast processing and real-time use cases.
- Graph databases represent relationships between entities.

Simple example

A business application may use a relational database for orders, object storage for invoices, a data warehouse for analytics, and a graph model for recommendation paths.

5.4 Networking Services

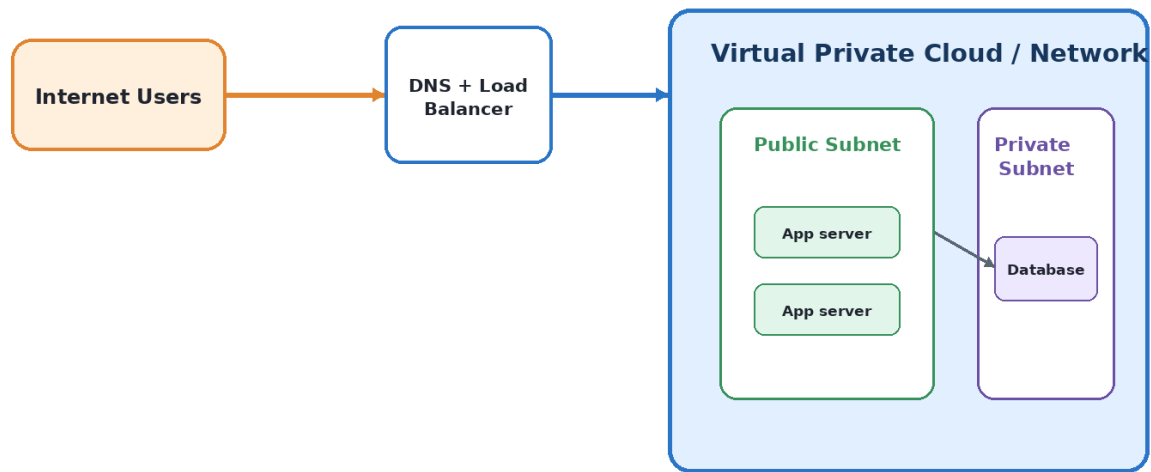
Learning outcome

Learners should understand the purpose of virtual networks, subnets, load balancers, DNS, VPNs, firewalls, and CDNs.

Networking services connect cloud resources securely and control how traffic flows. A typical cloud network separates public and private areas. Public-facing components such as load balancers receive user traffic, while databases remain private. Firewalls, security groups, route tables, VPNs, private links, and DNS help make cloud systems secure and reachable.

Cloud Networking Services

Connect resources securely and control traffic flow



Key concepts: VPC/VNet, subnet, route table, DNS, load balancer

Cloud Computing Fundamentals - Visual Tutorial

Figure 28: A typical cloud network separates public entry points from private backend resources.

Trainer explanation points

- Never place every resource directly on the public internet.
- Use load balancers to distribute traffic across healthy application instances.
- Use private subnets for databases and internal services.
- Use VPN or private connectivity to connect on-premise networks with cloud.

Simple example

A web app has a public load balancer, app servers in a controlled subnet, and a private database accessible only from application servers.

5.5 Identity and Access Management

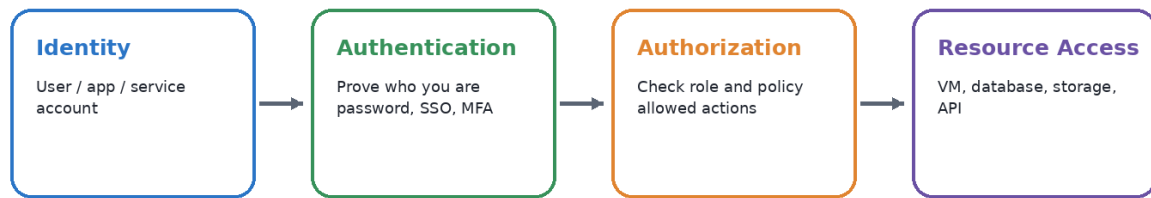
Learning outcome

Learners should understand how IAM controls cloud access.

Identity and Access Management controls who can access which resource and what action they can perform. IAM includes users, groups, roles, policies, permissions, service accounts, multi-factor authentication, and sometimes integration with corporate identity providers. IAM is one of the most important areas of cloud security.

Identity and Access Management (IAM)

Controls who can do what on which cloud resource



Best practice: Give minimum permissions needed, use roles instead of shared passwords, enable MFA, and review permissions regularly.

Cloud Computing Fundamentals - Visual Tutorial

Figure 29: IAM flow from identity to authentication, authorization, and resource access.

Trainer explanation points

- Authentication answers: Who are you?
- Authorization answers: What are you allowed to do?
- Least privilege means giving only the permissions needed for the job.
- Use roles and temporary credentials instead of shared passwords wherever possible.

Simple example

A developer may have permission to deploy applications in a development environment but not delete production databases.

5.6 Security Services

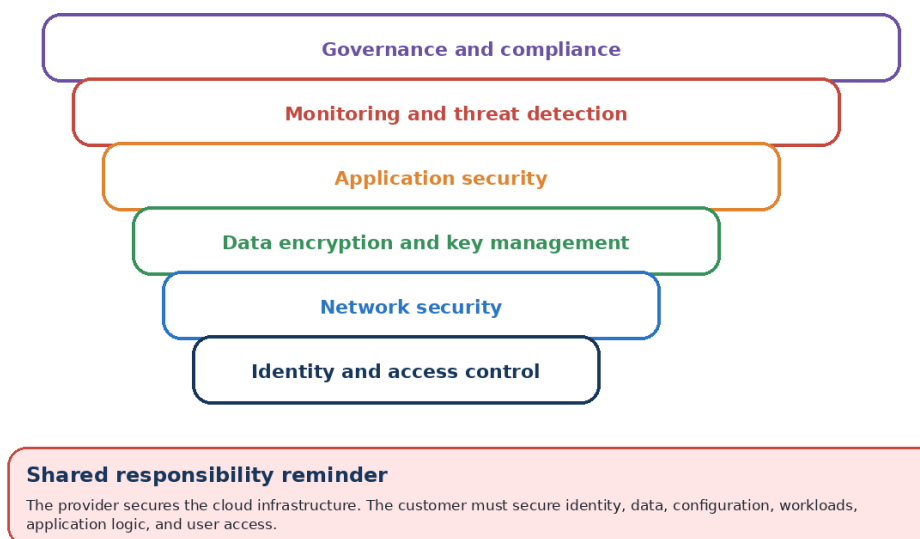
Learning outcome

Learners should understand cloud security as a layered model.

Cloud security is not a single service. It includes identity security, network security, data protection, encryption, key management, vulnerability management, threat detection, compliance, configuration management, and monitoring. Security must be designed from the beginning rather than added after deployment.

Cloud Security Overview

Security is layered: identity, network, data, application, monitoring, and governance



Cloud Computing Fundamentals - Visual Tutorial

Figure 30: Cloud security works through multiple layers of protection.

Trainer explanation points

- Use encryption for data at rest and in transit.
- Control access using IAM and multi-factor authentication.
- Use network rules to restrict traffic.
- Monitor suspicious activity and configuration changes.
- Follow the shared responsibility model carefully.

Simple example

A database may be encrypted, placed in a private subnet, accessible only to a backend role, backed up regularly, and monitored for unusual access.

5.7 Monitoring and Logging

Learning outcome

Learners should understand how monitoring supports reliability and troubleshooting.

Monitoring collects metrics, logs, traces, events, and alerts. Metrics show numerical values such as CPU usage, memory, response time, request count, and error rate. Logs explain what happened inside the application. Alerts notify teams when something goes wrong. Dashboards help teams see service health quickly.

Monitoring and Logging

Observe system health, detect issues, and improve reliability

Example: Response time crosses 3 seconds -> alert triggers -> team checks logs -> database query optimized.



Teaching point

Monitoring is not only for failures. It also helps with capacity planning, SLA reporting, security investigation, cost optimization, and performance tuning.

Cloud Computing Fundamentals - Visual Tutorial

Figure 31: Monitoring turns application signals into alerts, dashboards, and operational action.

Trainer explanation points

- Monitoring should be configured before production release.
- Logs help find root cause; metrics help detect symptoms.
- Alerts should be meaningful and actionable.
- Monitoring also supports security, cost management, capacity planning, and SLA reporting.

Simple example

If a website becomes slow, monitoring can show whether the cause is CPU overload, database delay, network issue, or application error.

5.8 Backup and Disaster Recovery

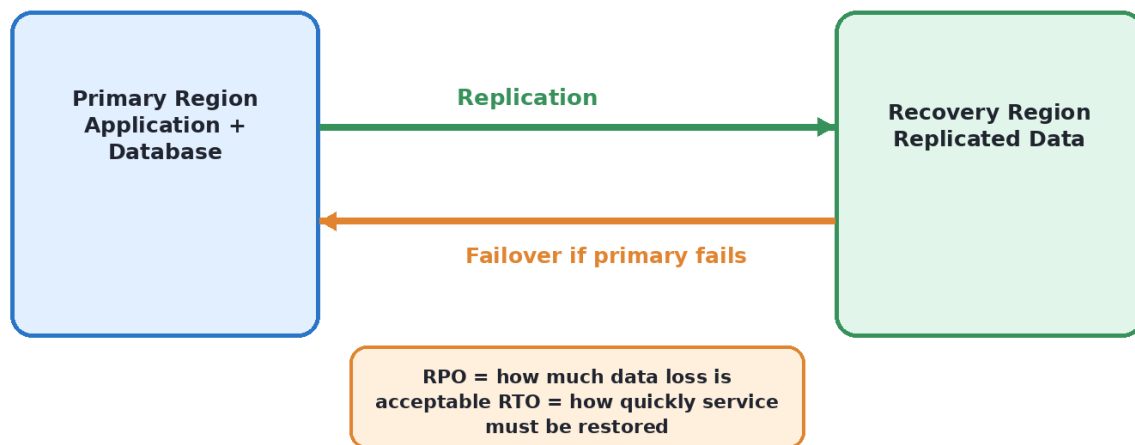
Learning outcome

Learners should understand backups, replication, RPO, RTO, and failover.

Backup and disaster recovery services protect systems against accidental deletion, hardware failure, cyberattack, region outage, and human error. Backup creates recoverable copies. Replication keeps data synchronized to another location. Disaster recovery defines how systems are restored after major failure.

Backup and Disaster Recovery

Prepare for failure before it happens



Cloud Computing Fundamentals - Visual Tutorial

Figure 32: Backup and disaster recovery planning uses replication, failover, RPO, and RTO.

Trainer explanation points

- RPO means Recovery Point Objective: how much data loss is acceptable.
- RTO means Recovery Time Objective: how quickly the service must be restored.
- High availability reduces downtime within normal operations.
- Disaster recovery prepares for larger failures.

Simple example

If RPO is 15 minutes, the business can tolerate losing up to 15 minutes of data. If RTO is 1 hour, the system must be restored within one hour.

5.9 Containers and Kubernetes Services

Learning outcome

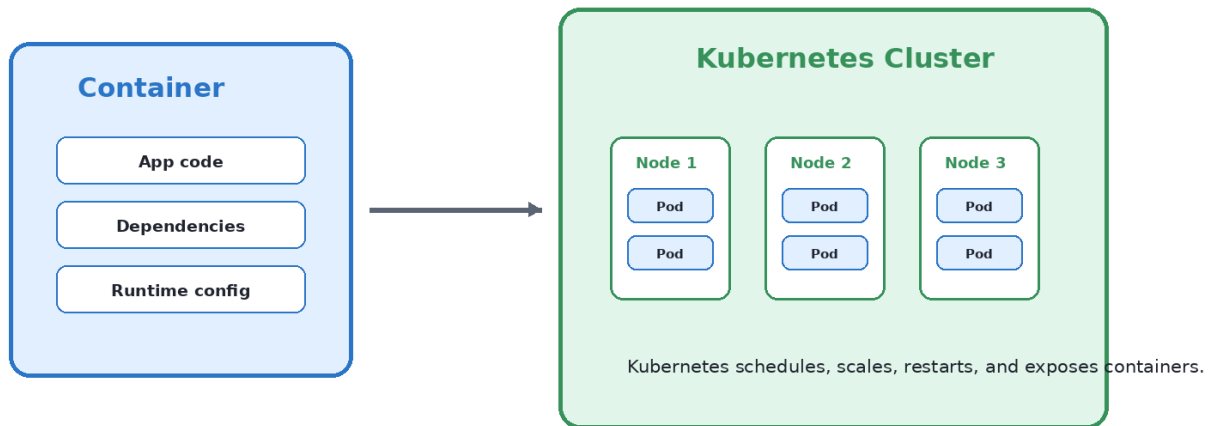
Learners should understand why containers and Kubernetes are used for modern applications.

A container packages application code and dependencies so it can run consistently across environments.

Kubernetes is a platform for running many containers. It can schedule containers across nodes, restart failed containers, scale workloads, and expose services. Kubernetes is popular for microservices but requires operational knowledge.

Containers and Kubernetes

Run portable applications and manage them at scale



Cloud Computing Fundamentals - Visual Tutorial

Figure 33: Containers package apps; Kubernetes orchestrates containers across nodes.

Trainer explanation points

- Containers improve portability between developer machines, test environments, and cloud.
- Kubernetes is useful when many containers need orchestration.
- For simple applications, PaaS may be easier than Kubernetes.
- Kubernetes still needs security, monitoring, upgrades, and governance.

Simple example

An online learning platform may run separate microservices for user profile, course catalog, video progress, payment, and certificate generation on Kubernetes.

5.10 Serverless Services

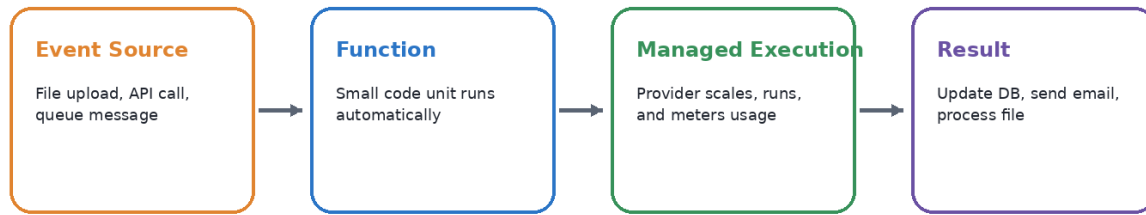
Learning outcome

Learners should understand serverless as a core cloud service family beyond the service model definition.

Serverless services include functions, event buses, managed workflows, queues, and serverless databases. The core pattern is event-driven execution. Serverless is useful for automation, integration, lightweight APIs, file processing, notifications, and scheduled jobs.

Serverless Event-Driven Flow

Code runs only when an event triggers it



Why it matters: no server provisioning, automatic scale, and cost is often tied to execution count/duration.

Cloud Computing Fundamentals - Visual Tutorial

Figure 34: Serverless services connect events to automatic execution and output actions.

Trainer explanation points

- Serverless can reduce operational overhead for event-driven workloads.
- Function permissions must be tightly controlled.
- Cold starts, execution duration limits, and observability should be considered.
- Serverless fits well with queues, storage events, APIs, and workflow services.

Simple example

A serverless function can generate a course completion certificate when a learner completes all modules.

5.11 AI, ML, Analytics, and Data Services

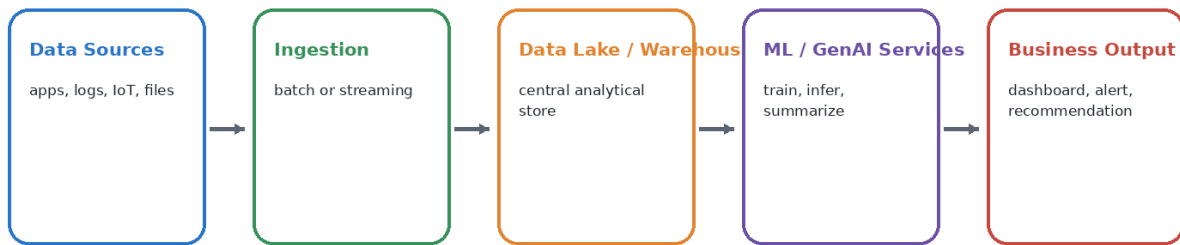
Learning outcome

Learners should understand how cloud platforms support modern data and AI use cases.

Modern cloud platforms provide services for data ingestion, data lakes, data warehouses, business intelligence, machine learning, generative AI, model deployment, speech, vision, document processing, and IoT analytics. These services help organizations convert raw data into insights, predictions, recommendations, and automated decisions.

AI, ML, Analytics, and Data Services

Cloud provides scalable data pipelines and ready AI capabilities



Example

A retail company ingests purchase data, analyzes behavior, trains recommendation models, and shows personalized product suggestions to customers.

Cloud Computing Fundamentals - Visual Tutorial

Figure 35: Cloud data and AI pipeline from sources to business output.

Trainer explanation points

- Data quality and governance are essential for AI success.
- AI services often depend on storage, database, networking, IAM, monitoring, and security.
- Analytics turns historical data into insights; ML predicts or classifies; generative AI summarizes, answers, drafts, or assists.

Simple example

A retail company uses cloud analytics to understand buying behavior and AI services to recommend products to customers.

5.12 Core Services Quick Reference Table

Service Family	What It Does	Common Examples	Beginner Analogy
Compute	Runs application logic and processing	VMs, containers, Kubernetes, PaaS, functions	Engine of the system
Storage	Stores files, objects, disks, backups, and archives	Object, block, file, archive storage	Warehouse for digital files
Database	Stores business records and queryable data	Relational, NoSQL, warehouse, graph, in-memory	Organized digital register
Networking	Connects services and controls traffic	VPC/VNet, subnets, DNS, load balancer, VPN, CDN	Roads and gates
IAM	Controls access and permissions	Users, roles, groups, policies, MFA	Security guard and permission card
Security	Protects resources, data, and users	Encryption, keys, firewall, threat detection	Locks, CCTV, rules
Monitoring	Tracks health and performance	Metrics, logs, traces, dashboards, alerts	Control room dashboard
Backup/DR	Restores systems after failure	Snapshots, replication, failover	Insurance and recovery plan
Containers/Kubernetes	Packages and orchestrates apps	Docker, managed Kubernetes	Shipping containers and port manager
AI/Analytics	Converts data into insights and intelligence	Data lake, BI, ML, GenAI	Business brain

Module 6: Real-World Cloud Architecture Example

This module connects all earlier concepts through one practical architecture. The goal is not to memorize product names. The goal is to understand how cloud service families work together.

6.1 Online Shopping Application Architecture

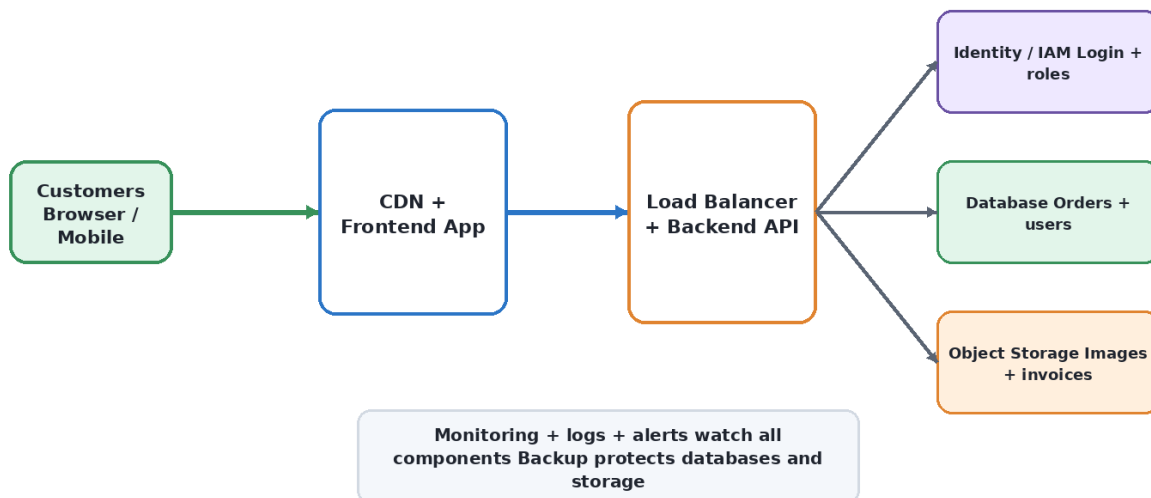
Learning outcome

Learners should map cloud concepts to a real application architecture.

An online shopping application needs a frontend, backend API, product images, user authentication, order database, payment integration, monitoring, backup, and security. Cloud services provide these building blocks in a scalable and managed way.

Reference Architecture: Online Shopping Application

A practical view of core cloud services working together



Cloud Computing Fundamentals - Visual Tutorial

Figure 36: Reference architecture for an online shopping application using core cloud services.

Trainer explanation points

- Users access the application through browser or mobile app.
- DNS and CDN help route users and deliver frontend content quickly.
- The backend API runs on compute or PaaS services.
- Product images and invoices are stored in object storage.
- Customer, product, and order data are stored in a database.
- IAM controls access; monitoring tracks health; backup protects data.

Simple example

When a customer opens a product page, the frontend loads from CDN, the API fetches product data from the database, product images load from object storage, and monitoring records performance.

6.2 Request Flow Explanation

1. Customer opens the shopping website from a browser or mobile device.
2. DNS resolves the application URL and traffic reaches the CDN or load balancer.
3. Frontend assets such as HTML, CSS, JavaScript, and images are served.
4. The frontend calls backend APIs for product, cart, order, or payment actions.
5. The backend API checks identity and permissions where required.
6. The API reads or writes data in the database.
7. Product images and invoices are stored or retrieved from object storage.
8. Monitoring collects metrics and logs. Alerts notify the operations team if thresholds are crossed.
9. Backups and replication protect important business data.
10. Security services protect access, traffic, secrets, and data.

Requirement	Cloud Service Mapping	Reason
Website frontend	CDN, static hosting, app service	Fast content delivery to users
Business logic	Backend API on VM, PaaS, container, or serverless	Processes product, cart, order, and payment requests
Product images	Object storage	Durable and scalable file storage
Orders and users	Relational database	Structured transactions and queries
Authentication	IAM / identity service	Secure login and role-based access
High traffic handling	Load balancer and auto-scaling	Distribute traffic and scale capacity
Security	Encryption, firewall, policies, secrets, monitoring	Protect system and data
Operations	Monitoring, logs, alerts	Troubleshooting and reliability
Recovery	Backup and replication	Restore after failure

Module 7: Mini Hands-On and Architecture Design

Activity

This activity can be done even without access to a cloud account. The objective is to make learners think like solution designers: identify requirements, select cloud services, and justify choices.

7.1 Final Activity: Design an Online Learning Platform

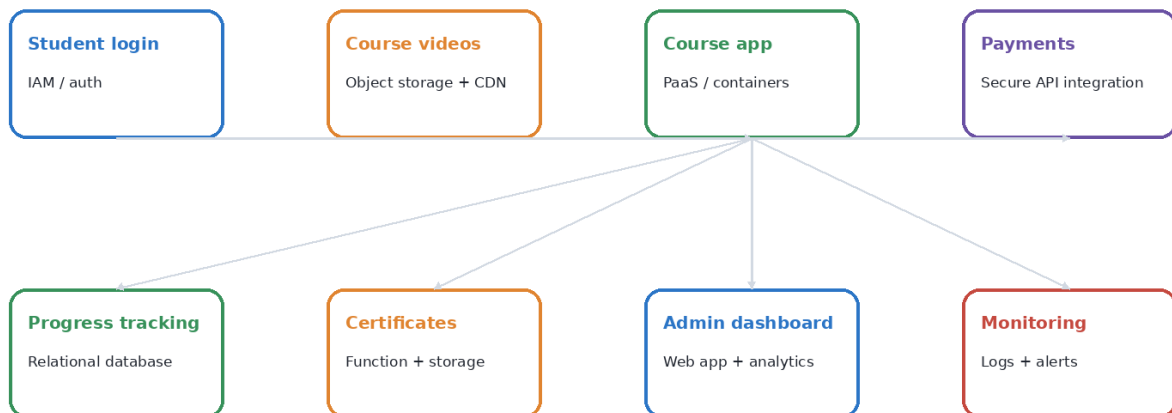
Learning outcome

Learners should apply cloud concepts by designing a basic architecture for a realistic platform.

The platform should support student login, course videos, payment, progress tracking, certificate generation, admin dashboard, reporting, monitoring, backup, and security. Learners should decide which cloud service family is needed for each requirement.

Final Activity Architecture: Online Learning Platform

Learners design this architecture as a group exercise



Exercise: ask learners to explain what happens when a student logs in, watches a video, completes a quiz, and downloads a certificate.

Cloud Computing Fundamentals - Visual Tutorial

Figure 37: Suggested architecture components for an online learning platform.

Trainer explanation points

- The point is not to choose exact AWS/Azure/GCP/SAP product names at first. First choose service families.
- Ask learners to justify why they selected compute, storage, database, IAM, CDN, monitoring, and backup.
- Encourage learners to explain traffic flow from login to video watching to certificate generation.

Simple example

A learner watches a course video. The user authenticates through IAM, video streams from object storage through CDN, progress is saved in a database, and monitoring tracks video delivery performance.

Mini activity

Divide learners into groups. Give them 15 minutes to create the architecture. Each group presents service choices in 3 minutes.

7.2 Activity Worksheet

Requirement	Recommended Service Family	Learner Notes
Student login and roles	IAM / authentication service	
Course video storage	Object storage + CDN	
Application backend	PaaS, containers, or serverless	
Payment processing	Secure API integration + secrets management	
Course progress tracking	Relational database or NoSQL database	
Certificate generation	Serverless function + object storage	
Admin dashboard	Web app + database + analytics	
Reports and analytics	Data warehouse / BI / analytics services	
Performance tracking	Monitoring, logs, metrics, alerts	
Data protection	Backup, replication, encryption	

7.3 Optional Cloud Console Demo

If a cloud account is available, the trainer can show a guided console walkthrough. The purpose is to make the concepts visible, not to teach one vendor in depth.

- Show the cloud dashboard and explain regions and accounts/subscriptions/projects.
- Open compute service and explain virtual machines, app hosting, or functions.
- Open storage service and explain buckets/containers and access settings.
- Open database service and explain managed database options.
- Open IAM and show users, roles, permissions, and policies.
- Open monitoring and show metrics, logs, dashboards, and alerts.
- Open billing/cost section and explain measured usage, budgets, and tags.

Safety note for demo

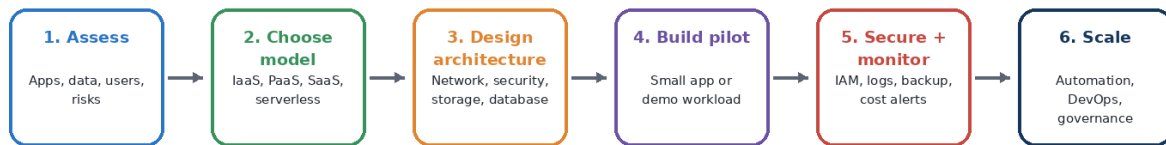
Do not create paid resources casually in a live class unless budget controls are configured. Prefer free-tier demos, screenshots, or architecture walkthroughs when the goal is conceptual learning.

Module 8: Recap, Quiz, and Q&A

End the tutorial by reinforcing the basic mental model: cloud is a way to consume technology services on demand. Service models decide responsibility boundaries. Deployment models decide where the cloud environment lives. Core services are the building blocks of real architectures.

Simple Cloud Adoption Roadmap

A beginner-friendly path from learning to implementation



Trainer reminder

Cloud adoption is not only technical. It also needs skills, security practices, cost ownership, governance, and change management.

Cloud Computing Fundamentals - Visual Tutorial

Figure 38: A simple learning-to-implementation cloud adoption roadmap.

8.1 Recap Points

- Cloud computing delivers IT resources over the internet on demand.
- Traditional IT focuses on owning and operating infrastructure; cloud focuses on consuming managed services.
- Key characteristics include on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service, automation, and global availability.
- IaaS provides infrastructure, PaaS provides a managed application platform, SaaS provides a complete application, and serverless runs event-driven code without server management.
- Public cloud is provider-owned shared infrastructure, private cloud is dedicated, hybrid cloud connects on-premise/private with public cloud, and multi-cloud uses more than one provider.
- Core services include compute, storage, databases, networking, IAM, security, monitoring, backup, disaster recovery, containers, Kubernetes, serverless, AI, ML, analytics, and data services.
- Good cloud architecture must include security, monitoring, backup, cost control, and governance from the beginning.

8.2 Quiz Questions with Answers

11. What is cloud computing?

Answer: Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, software, and platforms over the internet on demand.

12. What is the difference between traditional IT and cloud IT?

Answer: Traditional IT requires organizations to buy and maintain infrastructure. Cloud IT lets organizations consume resources from a provider and scale them as needed.

13. Which cloud characteristic allows resources to increase and decrease based on demand?

Answer: Rapid elasticity.

14. Which cloud characteristic supports pay-as-you-use billing?

Answer: Measured service.

15. Which service model gives virtual machines, storage, and networking?

Answer: IaaS.

16. Which service model allows developers to deploy code without managing servers and runtime manually?

Answer: PaaS.

17. Which service model provides complete software applications over the internet?

Answer: SaaS.

18. What is serverless computing?

Answer: Serverless computing allows code to run in response to events while the cloud provider manages server provisioning, scaling, and execution.

19. What is public cloud?

Answer: Public cloud is cloud infrastructure owned and operated by a third-party provider and shared securely among multiple customers.

20. What is hybrid cloud?

Answer: Hybrid cloud combines on-premise or private cloud environments with public cloud services.

21. What is IAM used for?

Answer: IAM controls users, roles, permissions, authentication, and authorization for cloud resources.

22. Why should a database usually be placed in a private subnet?

Answer: To reduce exposure to the public internet and allow access only from approved application components.

23. What is the difference between object storage and block storage?

Answer: Object storage stores files/objects with metadata and is good for images, logs, backups, and media. Block storage behaves like a disk attached to a VM.

24. What are RPO and RTO?

Answer: RPO defines acceptable data loss; RTO defines how quickly service must be restored after failure.

25. Why is monitoring important?

Answer: Monitoring helps detect issues, troubleshoot performance, trigger alerts, support security investigation, and improve reliability.

Appendix A: Trainer Script for Opening the Session

Good morning everyone. Today we will learn cloud computing from the foundation level. We will not start with product names. Instead, we will first understand the idea of cloud, then service models, deployment models, and finally the core services used to design real applications.

By the end of this session, you should be able to look at a business requirement and identify which cloud services are needed. You should also understand the difference between IaaS, PaaS, SaaS, serverless, public cloud, private cloud, hybrid cloud, and multi-cloud.

The most important message is this: cloud computing is not only about renting servers. It is about speed, flexibility, scalability, automation, global reach, and business innovation. But cloud also needs security, governance, cost control, monitoring, and responsible design.

Appendix B: Glossary of Cloud Terms

Term	Meaning
Account / Subscription / Project	A logical container in a cloud provider where services, billing, permissions, and resources are managed.
Region	A geographic area where cloud data centers are located.
Availability Zone	An isolated location within a region designed for high availability.
Virtual Machine	A software-based server running on cloud infrastructure.
Container	A packaged application unit containing code and dependencies.
Kubernetes	A container orchestration platform for managing containerized workloads.
Object Storage	Storage for files/objects such as images, backups, logs, videos, and documents.
VPC / VNet	A private virtual network in the cloud.
Subnet	A smaller network segment inside a virtual network.
Load Balancer	A service that distributes traffic across multiple healthy servers or instances.
IAM	Identity and Access Management: users, roles, permissions, and access control.
MFA	Multi-factor authentication: additional verification beyond password.
Auto-scaling	Automatic increase or decrease of resources based on demand.
RPO	Recovery Point Objective: acceptable amount of data loss.
RTO	Recovery Time Objective: acceptable time to restore service.
CDN	Content Delivery Network: distributes content closer to users for faster access.
API	Application Programming Interface: a controlled way for software systems to communicate.
DevOps	Practices that combine development and operations through automation, CI/CD, and collaboration.

Appendix C: Recommended Next Learning Path

After this 4-hour tutorial, learners can continue with more specific cloud and platform topics.

- Cloud networking fundamentals: VPC/VNet, subnets, routing, DNS, load balancing, VPN, CDN.
- Linux and virtual machines for cloud administration.
- Cloud storage and database design basics.
- Identity and access management and cloud security fundamentals.

- Cloud cost optimization: tagging, budgets, right-sizing, reserved capacity, storage tiering.
- DevOps and CI/CD pipeline basics.
- Containers and Kubernetes fundamentals.
- Serverless application design.
- Data analytics and AI services on cloud.
- Platform-specific paths: AWS Cloud Practitioner, Microsoft Azure Fundamentals AZ-900, Google Cloud Digital Leader, SAP BTP Fundamentals.

Appendix D: One-Page Cloud Architecture Checklist

Checklist Area	Questions to Ask
Business goal	What problem is the application solving? Who are the users? What is the expected traffic?
Compute	Will the workload run on VM, PaaS, container, Kubernetes, or serverless?
Storage	What files or objects are stored? How often are they accessed? What retention is required?
Database	What data model is needed: relational, NoSQL, analytics, graph, or in-memory?
Networking	Which resources are public? Which must stay private? Is VPN/private link needed?
Identity	Who can access the system? What roles and permissions are required?
Security	Is data encrypted? Are secrets protected? Are firewalls and policies configured?
Monitoring	What metrics, logs, dashboards, and alerts are required?
Backup/DR	What are RPO and RTO? What backup and restore process is required?
Cost	Are tags, budgets, alerts, and right-sizing practices in place?
Governance	Who owns the environment, access, cost, compliance, and operational support?

Closing message

Cloud computing is the foundation of modern digital systems. Once learners understand the basic models and core service families, they can learn any specific cloud platform much faster because the underlying concepts remain similar.